

- Two vectors of equal magnitudes of 20 units are given in x-y plane as shown in the figure below. (1)
What are the magnitude and the direction of the resultant of the two vectors?
[Figure: A Cartesian coordinate system with horizontal x-axis and vertical y-axis. Two vectors of magnitude 20 units originate from the origin: one vector is in the first quadrant labeled '20 units' at an angle of 30° counterclockwise from the positive x-axis; the second vector is in the second quadrant labeled '20 units' at an angle of 150° counterclockwise from the positive x-axis.]
A. 20.00 units along positive y- axis
B. 40.00 units along positive y- axis
C. 34.40 units along negative x- axis
D. 34.40 along positive x-axis
- A driver of an automobile, moving with 30 m/s, applies a break when he observes a stopping light 180 m ahead. (1)
If the automobile decelerates uniformly and comes to stop just at the mark of the stopping light, what is the magnitude of deceleration of the automobile?
A. 2.5 m/s^2
B. 2.0 m/s^2
C. 12.0 m/s^2
D. 1.5 m/s^2
- Consider the following three phrases, (1)
I. The acceleration of an object at specific time
II. The ratio of change in velocity to significant value of time interval Δt
III. The ratio of change in velocity to time interval Δt where Δt approaches to zero
Which one of the following statements is correct about the expressions found in the given phrases?
A. I and II express average acceleration whereas III expresses instantaneous acceleration.
B. III expresses average acceleration whereas I and II express instantaneous acceleration.
C. II expresses average acceleration whereas I and III express instantaneous acceleration.
D. I and III express average acceleration whereas II expresses instantaneous acceleration.
- When a constant force of magnitude F is applied on an object, it produces a magnitude of acceleration, a. By (1)
what factor does the acceleration change if the magnitude of the force is doubled and the mass of the object is halved?
A. 4
B. $\frac{1}{4}$
C. 2
D. $\frac{1}{2}$

5. Assuming all the given curves are parabolic and the motion right direction is positive, which one of the following graphs represent the displacement-time graph of an object moving to the left with a constant acceleration towards the right? (1)

[Figure: Four displacement-time graphs, each plotting displacement $\vec{s}(\text{m})$ on the vertical axis against time $t(\text{s})$ on the horizontal axis, starting from the origin: - Graph A: Located in the first quadrant, showing a parabolic curve starting at the origin with zero initial slope and curving upward with increasing positive slope (concave upward). - Graph B: Located in the fourth quadrant, showing a parabolic curve starting at the origin with zero initial slope and curving downward with increasingly negative slope (concave downward). - Graph C: Located in the first quadrant, showing a parabolic curve starting at the origin with a positive initial slope that flattens out (decreases towards zero) over time (concave downward). - Graph D: Located in the fourth quadrant, showing a parabolic curve starting at the origin with a steep negative initial slope that curves upward and flattens out towards zero slope over time (concave upward).]

- A.** [Graph A] **B.** [Graph B]
C. [Graph C] **D.** [Graph D]

6. If the weight of an object on the surface of the earth is 2000 N, what is the weight of the same object on the surface of another planet with gravitational acceleration of 5 m/s^2 ? (Acceleration due to gravity on the Earth is, $g = 10 \text{ m/s}^2$)

- A. 2000 N
B. 1000 N
C. 40 N
D. 20 N

7. A block of mass $m = 30 \text{ kg}$ is released from rest at point A on a frictionless surface as shown in the figure below. (1)

What is the speed of the block when it arrives at point B? ($g = 10 \text{ m/s}^2$)

[Figure: A curved frictionless track profile is shown above a horizontal reference ground line. At the starting point A, the track is at a height of 50 m above the ground, where a block is released from rest. The track curves down into a valley, rises over an intermediate hill, descends into another valley, and rises to a peak labeled point B at a height of 30 m above the ground, before sloping downward again. A horizontal distance of 150 m is indicated along the bottom between the vertical projection of point A and point B.]

- A. 100 m/s
B. $10\sqrt{10}$ m/s
C. 15 m/s
D. 20 m/s

8. Suppose that you repeatedly observe the electric 'mitad' at your home heats up slower than that of your neighbor when connected to the same source while cooking 'injera'. As compared to your neighbor's, your 'mitad' is

- A. made up of higher specific heat material.
- B. made up of higher conductivity material.
- C. made up of lower mass rigid material.
- D. allowed to get lower energy than the source.

9. In determining the specific heat capacity of a solid by electrical method, which of the following process or assumptions is possible? (1)

- A. Allowing heat flow to the environment.
- B. Considering heat flow to the thermometer.
- C. Assuming the electrical heater is 100 % efficient.
- D. Assuming heat flows from the solid object to electrical heater.

- 10.** Two charges $q_1 = +2 \mu\text{C}$ and $q_2 = +8 \mu\text{C}$ are placed along the x-axis as shown in the figure below. Where should a charge q_3 be placed along the x-axis for the resultant electric force on it to be zero? [Figure: A horizontal axis labeled 'x-axis' with an arrow pointing to the right. Two point charges are positioned on the axis: the left charge is labeled $q_1 = +2 \mu\text{C}$ and the right charge is labeled $q_2 = +8 \mu\text{C}$. A dimension line indicates a distance of 6 m between q_1 and q_2 . A point between them along the axis is marked and labeled q_3 .] (1)

 - 2 m from q_1
 - 2.5 m from q_1
 - 4 m from q_1
 - 5 m from q_1

11. A measure of number of electric field lines that crosses a given area is (1)

 - electric field.
 - electric flux.
 - electric force.
 - electric potential.

12. Which one of the following is correct about elastic and inelastic collisions? (1)

 - Total kinetic energy of the system is conserved in both elastic and inelastic collisions.
 - Total kinetic energy of the system is conserved in an elastic collision but not in an inelastic collision.
 - Total linear momentum of the system is conserved in an elastic collision but not in an inelastic collision.
 - Momentum and kinetic energy of the system are conserved if colliding objects stick together after collision.

13. Two point charges are fixed on the x-axis, in such a way that $8 \mu\text{C}$ is placed at $x = 0$ while $-10 \mu\text{C}$ at $x = 9.0 \text{ cm}$. Taking the electric potential at infinity being zero, at what point on the x-axis between the two charges is the potential equal to zero? (1)

 - $x = 9.0 \text{ cm}$
 - $x = 5.0 \text{ cm}$
 - $x = 4.5 \text{ cm}$
 - $x = 4.0 \text{ cm}$

14. Which one of the following statements is correct about equipotential surface? (1)

 - The electric potentials at different points on a line in a uniform electric field region are the same, if the line is perpendicular to the electric field.
 - The electric potentials at different points on the line in a uniform electric field region are different, if the line is perpendicular to the electric field.
 - The magnitudes of electric potentials on each surface of two different concentric circles centered at a point charge are the same.
 - The magnitudes of electric potentials at different points on the same circle centered at a point charge are different.

15. In conversion of galvanometer to voltmeter, we connect (1)

 - a low resistance resistor in parallel with the galvanometer.
 - a high resistance resistor in parallel with the galvanometer.
 - a low resistance resistor in series with the galvanometer.
 - a high resistance resistor in series with the galvanometer.

16. One of the following is NOT correct about electric resistance of a conductor. (1)

 - The resistance can be due to the collision between free electrons.
 - The resistance can be due to the collision between free electron and
 - The resistance of a conductor can increase because of a large free electrons.
 - Different substances can have different electric current resistance.

17. As shown in the figure three resistors are connected to a battery of emf $\varepsilon = 12 \text{ V}$. (1)
How much is the current in 2Ω resistor?
[Figure: A circuit diagram with three parallel branches connected between a top node and a bottom wire. The central branch contains a resistor labeled " 2Ω " in series above a DC battery labeled " 12 V " (with the longer positive terminal on top). Connected in parallel to this central branch are two outer branches: the left branch has a resistor labeled " 3Ω " in its top segment, and the right branch has a resistor labeled " 6Ω " in its top segment. Both outer branches connect back to the bottom of the central branch via a common bottom wire.]
- A. 12.0 A B. 3.0 A
C. 6.0 A D. 2.4 A
18. When a dielectric material is inserted between an isolated charged parallel plate capacitor, which of the following effect happens due to the polarization of the charge? (1)
- A. The potential difference between the plates remains constant.
B. The charge stored on the plates is increased.
C. The electric field between the plates decreases.
D. The capacitance of the capacitor decreases.
19. In nuclear experiment, if the fired atomic bullet collides with a nucleus of a given element and produces a new kind of particles as massive as proton which penetrates through several inches of lead, which of the following sub-atomic particle is found by this experiment? (1)
- A. Proton B. Electron
C. Nucleon D. Neutron
20. Which one of the following process increases the nuclear stability? (1)
- A. Moving the system to higher energy state.
B. Moving the system to the lower energy state.
C. Decreasing the binding energy of the nucleus.
D. Increasing the disruptive energy of the nucleus.
21. The reason why carbon-14 dating is not applicable in dating of old biological materials is (1)
- A. its abundance is too large to count in old biological materials.
B. it has different chemistry from stable carbon.
C. its concentration falls nearly zero in old biological materials.
D. it is difficult to compare the existing carbon-14 in the artifact with the normal abundance in living tissue.
22. What is the expression of a half-life of a radioactive nucleus with decay constant λ subject to the decay law? (1)
- A. $\frac{0.693}{\lambda}$ B. $\frac{\lambda}{0.693}$
C. $\frac{2 \ln 2}{\lambda}$ D. $\frac{\lambda}{2}$

— END OF EXAMINATION —